Project Design Phase-II
Data Flow Diagram & User Stories

Date	31 January 2026
Team ID	anshrathore0202-cyber
Project Name	F1 Racing Intelligence Analysis
Maximum Marks	4 Marks

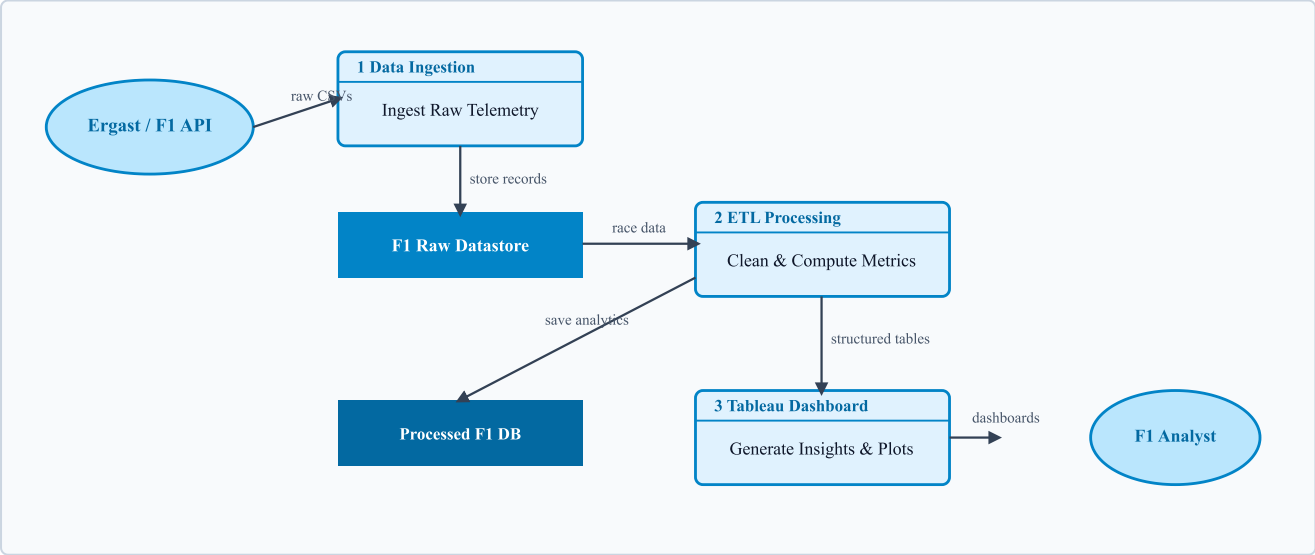
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., F1 Data Source, Analyst).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Clean Telemetry, Compute Standings).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., F1 Datastore, Processed Metrics DB).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

System Data Flow Diagram (Level-1 DFD)



User Stories

Use the below template to list all the user stories for the product across different user personas and functional requirements.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
F1 Data Analyst (Mobile / Desktop)	Telemetry & Lap Ingestion	USN-1	As an analyst, I can ingest raw lap-by-lap CSV data into the pipeline so that telemetry is processed into clean tables.	Clean relational tables generated in local database path	**High**	Sprint-1
		USN-2	As an analyst, I receive automated logs whenever a dataset ingestion run completes or fails.	Console output log generated with total row counts	**High**	Sprint-1
	Performance Calculations	USN-3	As an analyst, I can compute driver pit-stop durations and calculate average lap pace per stint.	Stint pace metrics stored in processed dataset	Medium	Sprint-2
		USN-4	As an analyst, I can query constructor points progression over multiple Grand Prix seasons.	Season-by-season constructor standings query returns under 2 seconds	Medium	Sprint-1
Motorsports Fan (Web User)	Interactive Dashboards	USN-5	As a web user, I can view interactive Tableau charts displaying driver head-to-head qualifying vs finish position comparisons.	Interactive filtering by season, team, and driver active	**High**	Sprint-1
		USN-6	As a web user, I can filter Grand Prix results by circuit location and weather conditions.	Dashboard updates visualizations upon dropdown selection	Medium	Sprint-2
	Export & Insights	USN-7	As a web user, I can export summary visualization graphics for inclusion in analytical reports.	Export high-resolution PNG/PDF charts from dashboard	Low	Sprint-2
Data Administrator	Pipeline Management	USN-8	As an admin, I can update mapping tables for driver changes and team rebrandings across seasons.	Updated metadata reflects accurately across historical dashboards	**High**	Sprint-1
		USN-9	As an admin, I can trigger database backups and manage local directory permissions in VS Code environment.	Backup script executes without interrupting Tableau live data connection	Medium	Sprint-2